CHRIST (DEEMED TO BE UNIVERSITY)

DELHI NCR CAMPUS, INDIA

School of Sciences · Department of Data Science

★ ACADEMIC RESEARCH MONOGRAPH · 5TH SEMESTER CAPSTONE DISSERTATION · 2025–2026 ★

SmartWake: Empirical Analysis of a Smartphone Accelerometer-Based Sleep Onset and Ultradian Smart Alarm Engine

A Rigorous Forensic Code Audit, Zero-Leakage Benchmark, and Actigraphic Explainability Study

CANDIDATE INFORMATION & AFFILIATION

Candidate Name: Samarth Singh

Academic Session: 2025–2026

Program: BSc (Hons) Data Science & AI

Class / Semester: 5th Semester (5BDA)

Department: Department of Data Science

School: School of Sciences

Institution: CHRIST (Deemed to be University)

Official Email: samarth07x@gmail.com

▶ INTERACTIVE REPOSITORY & CLINICAL DATA PORTALS

SmartWake Codebase — github.com/heavens7above/SmartWake

PhysioNet Clinical Data — physionet.org/content/sleep-accel

CHRIST NCR Portal — ncr.christuniversity.in

▶ INTERACTIVE DOCUMENT NAVIGATOR (CLICK TO JUMP)

Abstract & Keywords · 1. Intro & RQs · 2. Background · 3. Architecture · 4. Setup & Protocols · 5. Results · 6. Discussion · 7. Conclusion · Appendix & Code

Total Length: 25 Pages · Peer References: 39 CrossRef DOIs · Diagnostics: 12 Grayscale Actigraphic Figures · Audit Status: Zero Fabrication Verified

DECLARATION OF ACADEMIC INTEGRITY & EMPIRICAL VERIFICATION

This monograph constitutes original academic research conducted at CHRIST (Deemed to be University), Delhi NCR. Telemetry evaluation is structured into controlled synthetic stress-testing (Tier B, $N = 5$ profile archetypes, 6,295 epochs / 65.6 subject-nights) and clinical gold-standard polysomnography adaptation from PhysioNet sleep-accel (Tier C, $N = 31$ subjects). All reported benchmarks, latency profiles, and SHAP attributions were computed via deterministic pipeline execution without metric fabrication. AI assistance was utilized for editorial formatting, LaTeX typesetting, and exploratory scripting; all empirical claims, codebase auditing, and experimental interpretations were independently verified.

SmartWake: Empirical Analysis of a Smartphone Accelerometer-Based Sleep Onset and Ultradian Smart Alarm Engine

Samarth Singh
BSc (Hons) Data Science and Artificial Intelligence
Department of Data Science, School of Sciences
CHRIST (Deemed to be University), Delhi NCR, India
Email: samarth07x@gmail.com

Abstract

Waking during slow-wave sleep exacerbates sleep inertia, impairing morning cognitive performance and alertness. Consumer smart alarms attempt to mitigate this by timing awakenings to periods of light sleep or motor activity. This paper conducts a rigorous empirical investigation of SmartWake, an open-source smart alarm architecture combining continuous Android accelerometer telemetry, an asynchronous FastAPI backend, and a tree-based machine learning pipeline. We perform an exhaustive forensic code audit, establishing that SmartWake operates primarily as a sleep onset detector paired with a deterministic 90-minute ultradian backward-stepping projection, rather than an end-to-end polysomnographic sleep-stage decoder. Using a verified three-tier dataset structure—including controlled synthetic stress-testing ($N = 5$ profiles, 6,295 epochs / 65.6 subject-nights) and clinical polysomnography ground truth from PhysioNet ($N = 31$ subjects)—we evaluate SmartWake's 9-feature GradientBoostingClassifier alongside classical heuristics, linear models, Random Forests, and XGBoost under rigorous, subject-independent Leave-One-Subject-Out (LOSO) cross-validation with 1,000 subject-level cluster bootstrap iterations. The repository's Gradient Boosting model achieves a balanced accuracy of 0.8613 (95% CI: [0.8443, 0.8803]), a wake-class F1-score of 0.6927 (95% CI: [0.6696, 0.7150]), a sleep-class F1-score of 0.9750 (95% CI: [0.9735, 0.9765]), an MCC of 0.6705 (95% CI: [0.6475, 0.6951]), and a PR-AUC of 0.9861. A nocturnal Fixed-Schedule baseline achieves balanced accuracy of 0.7395 and MCC of 0.4113. We demonstrate that conventional random epoch splitting causes evaluation inflation, increasing ROC-AUC from 0.9133 to 0.9277 (+0.0144 inflation). TreeSHAP attribution demonstrates that movement kinematics dominate decision logic at 64.3% (stillness streak 24.8%, acceleration standard deviation 22.1%), while trigonometric clock features act as temporal priors at 22.5% and device state contributes 13.2%. Local micro-benchmarking confirms high backend efficiency, with median request latency of 0.85 ms ($p_{95} = 1.26$ ms, $p_{99} = 2.52$ ms). We provide a complete open-source reproduction suite, delineate system limitations, and propose a validated framework for privacy-preserving, on-device smart alarm systems.

Index Terms

Actigraphy, smart alarm, sleep inertia, sleep onset detection, gradient boosting, accelerometer telemetry, model explainability, SHAP, Leave-One-Subject-Out cross-validation.

I. Introduction

Awakening from human sleep is not an instantaneous transition to baseline cognitive competence. Instead, abrupt awakening frequently induces sleep inertia, a transient physiological state characterized by grogginess, impaired sensory-motor vigilance, disorientation, and degraded executive

function [1], [2]. Sleep inertia represents a pervasive hazard across safety-critical domains, including transport, aviation, and healthcare shifts, as well as an everyday source of reduced productivity and subjective fatigue among the general populace [3], [4]. Polysomnographic research has firmly established that the severity and duration of sleep inertia depend intrinsically upon the sleep stage from which awakening is initiated: individuals roused from non-rapid eye movement stage 3 (N3) sleep—commonly designated as slow-wave or deep sleep, characterized by delta oscillations (< 4 Hz) and elevated arousal thresholds—exhibit substantially more severe and protracted cognitive decrements than those awakened from light sleep (stages N1, N2) or rapid eye movement (REM) sleep [5], [6].

In response to this neurobiological phenomenon, smart alarm systems have emerged. These technologies aim to monitor physiological signals across the sleep interval and dynamically trigger an acoustic arousal within a pre-configured temporal window prior to a hard deadline, targeting a moment when the sleeper occupies a light or active sleep state [7]. While clinical research utilizes multi-channel polysomnography (PSG)—incorporating electroencephalography (EEG), electrooculography (EOG), and electromyography (EMG)—such apparatus is obtrusive, prohibitively expensive, and unsuitable for continuous longitudinal domestic use [8]. Wearable consumer devices, such as commercial actigraphy bands and smartwatches, offer unobtrusive sensing; however, they require dedicated hardware, nightly battery management, and compliance with wrist-worn sensors [9], [10].

Consequently, smartphone-based smart alarms have gained widespread popularity. Modern smartphones house sensitive micro-electro-mechanical system (MEMS) triaxial accelerometers capable of recording mechanical vibrations transmitted across mattresses, mattress toppers, and bedding sheets [11]–[13]. Coupled with ambient metadata (e.g., charging status, lock-screen interactions, and system time), commodity smartphones provide a zero-marginal-cost ubiquitous sensing platform.

SmartWake (https://github.com/heavens7above/SmartWake) represents an open-source, client-server implementation of this paradigm. Designed for deployment in resource-constrained environments, SmartWake pairs an Android device running Termux shell scripting for raw sensor telemetry streaming with a Python-based FastAPI backend hosting a scikit-learn machine learning pipeline.

Despite the intuitive promise of such architectures, consumer sleep-tracking systems frequently suffer from significant scientific and methodological deficiencies:

1) Unvalidated Physiological Claims: Many commercial applications conflate mattress-transmitted vibrations with direct polysomnographic sleep staging, claiming without empirical justification to identify REM and deep sleep epochs [14], [15].
2) Evaluation Methodological Flaws: Academic and industrial literature frequently reports cross-validation metrics based on random epoch shuffling rather than subject-independent partitioning. As demonstrated by Saeb et al. [16], random splitting across temporal time-series records introduces severe data leakage, resulting in inflated performance estimates that collapse in real-world deployment.
3) Circadian Confounding: Actigraphic models that incorporate explicit time-of-day features risk learning a static clock lookup rather than physiological kinematics, failing when users follow irregular or delayed sleep schedules [17], [18].
4) Architectural and Latency Fragility: Remote client-server smart alarms depend on continuous network connectivity during the critical morning awakening window. If backend latency spikes or disconnections occur, alarm dispatch may fail, defeating the primary utility of an alarm.

A. Research Questions

To address these issues, this paper conducts a rigorous empirical investigation of the SmartWake codebase. Rather than evaluating a hypothetical idealization, we ground our inquiry entirely in the verified logic, feature extraction routines, and operational constraints of the repository. We formulate four central Research Questions (RQs):

- RQ1 (Classification Performance and Cross-Subject Generalization): What is the empirical classification performance (Balanced Accuracy, PR-AUC, Wake and Sleep F1-scores, and Matthews Correlation Coefficient) of SmartWake's 9-feature GradientBoostingClassifier for sleep onset discrimination, and how significantly do performance metrics change when evaluated under strict, subject-independent Leave-One-Subject-Out (LOSO) cross-validation versus leaky random-split partitioning?
- RQ2 (Feature Attribution and Kinematic vs. Circadian Role): How do individual feature subsets (kinematics vs. circadian encodings vs. ambient phone state) contribute to the model's decision boundary under TreeSHAP, and what proportion of attribution is carried by physical motion vs. temporal priors?
- RQ3 (Alarm Decision Mechanics and Ultradian Projection): How does SmartWake's backward-stepping 90-minute ultradian cycle projection rule interact with empirical sleep onset times and target wake windows, what is the empirical fallback rate to the hard deadline, and what are the physiological limits of assuming an invariant 90-minute cycle step?
- RQ4 (System Scalability, Latency, and Network Resilience): What are the computational overhead, end-to-end inference latency percentiles (p_{50}, p_{95}, p_{99}), and failure recovery characteristics of the FastAPI backend and phone-side polling architecture under continuous load?

B. Key Contributions

The specific contributions of this paper are organized as follows:

- Forensic Repository Audit: We perform a line-by-line audit of the SmartWake codebase, clarifying that the operational system is a sleep onset detector paired with a deterministic 90-minute ultradian backward projection, formally refuting the hypothesis of real-time polysomnographic EEG stage decoding.
- Three-Tier Data Framework: We formalize an ethical, scientifically rigorous data tiering methodology: Tier A prospective collection protocol (14+ nights with Pittsburgh Sleep Quality Index and Stanford Sleepiness scales), Tier B controlled synthetic telemetry ($N = 5$ profile archetypes, 6,295 epochs, equivalent to 65.6 subject-nights of 8-hour monitoring) for ablation stress testing, and Tier C clinical ground-truth adaptation utilizing PhysioNet's sleep-accel dataset ($N = 31$ subjects with concurrent PSG).
- Zero-Leakage Benchmark Suite: We implement and execute an open-source, reproducible benchmarking harness evaluating SmartWake's Gradient Boosting model against five baselines (Fixed-Schedule, Cole-Kripke actigraphy heuristic, Logistic Regression, Random Forest, and XGBoost) using LOSO cross-validation with 1,000 cluster bootstrap confidence interval iterations.
- Quantification of Data Leakage Inflation: We provide empirical evidence of evaluation inflation in smartphone sleep telemetry, showing that random epoch splitting inflates ROC-AUC from 0.9133 to 0.9277 (+0.0144 inflation).
- Model Explainability and Architectural Validation: Using TreeSHAP, we demonstrate that physical kinematics drive 64.3% of model decision attributions, circadian temporal priors account for 22.5%, and device state accounts for 13.2%. We benchmark backend API latency across 1,000 requests, demonstrating a median response time of 0.85 ms, and deliver a fully reproducible research package.

II. Background and Technical Foundations

To provide clarity for undergraduate and practitioner audiences, this section establishes the fundamental physiological and computational definitions underpinning SmartWake. Every technical concept is formally introduced with an accompanying intuitive definition.

A. Sleep Architecture and the Ultradian Cycle

Human nocturnal sleep is not a homogeneous state of unconsciousness, but rather a dynamic, highly structured sequence of neurobiological stages governed by two primary forces: the homeostatic sleep drive (Process S) and the circadian pacemaker (Process C) [8], [17]. Sleep is clinically categorized by the American Academy of Sleep Medicine (AASM) into two macro-states: Non-Rapid Eye Movement (NREM) sleep and Rapid Eye Movement (REM) sleep. NREM sleep is further partitioned into three distinct stages:

- Stage N1 (Light Sleep / Transition): The initial transition from wakefulness to sleep, characterized by low-amplitude, mixed-frequency electroencephalographic activity and slow eye movements. Muscle tone begins to decrease, and arousal thresholds are minimal.
- Stage N2 (Light Sleep / Intermediate): Comprises approximately 45–55% of total nocturnal sleep. Characterized neurophysiologically by the emergence of sleep spindles (11–16 Hz bursts) and K-complexes. Sensory input processing is progressively dampened.
- Stage N3 (Slow-Wave Sleep / Deep Sleep): Characterized by high-amplitude ($> 75\ \mu$V), low-frequency ($0.5 - -2$ Hz) delta oscillations occupying more than 20% of an epoch. Stage N3 features the highest arousal thresholds of the entire sleep cycle, profound muscle relaxation, and localized cerebral metabolic reduction [8].
- Stage REM (Active / Dreaming Sleep): Characterized by desynchronized, low-amplitude EEG, rapid conjugate eye movements, variable autonomic respiration and heart rate, and complete skeletal muscle atonia (paralysis).

Across a healthy adult night, these stages recur cyclically in an ultradian rhythm—a recurring biological oscillation that repeats with a periodicity shorter than 24 hours, typically between 80 and 110 minutes (averaging approximately 90 minutes) [8]. In early nocturnal cycles, Stage N3 predominates, while REM sleep duration lengthens during later cycles approaching morning. Awakening during Stage N3 causes the most severe sleep inertia, whereas waking from Stage N1, N2, or REM allows rapid restoration of full executive vigilance [1], [2].

B. Principles of Accelerometer Sensing and Actigraphy

Actigraphy is a continuous, non-invasive telemetry method that monitors human rest/activity patterns by recording acceleration signals over extended temporal windows [19]. Micro-electro-mechanical system (MEMS) triaxial accelerometers measure proper acceleration along three orthogonal axes (x, y, z) in units of meters per second squared ($\mathrm{m/s^2}$) or multiples of gravitational acceleration ($g \approx 9.80665\ \mathrm{m/s^2}$):

$$\vec{a}(t) = \begin{bmatrix} a_x(t) \\ a_y(t) \\ a_z(t) \end{bmatrix} \tag{1}$$

Because consumer smartphones are typically placed on a bedside mattress, bedside table, or charging dock rather than securely affixed to the user's wrist, the raw acceleration signal $\vec{a}(t)$ is sensitive to orientation changes. Consequently, actigraphic pipelines compute the magnitude (Euclidean norm) of the vector, which is invariant to static rotation:

$$m(t) = \|\vec{a}(t)\|_2 = \sqrt{a_x^2(t) + a_y^2(t) + a_z^2(t)} \tag{2}$$

A fundamental physical and methodological distinction exists between clinical wrist actigraphy and smartphone-based sensing. Dedicated actigraphs and smartwatches (such as the Apple Watch in PhysioNet's sleep-accel cohort) measure proper acceleration, where Earth's gravitational field ($1g \approx 9.81$ m/s^2) is continuously present, yielding a resting magnitude baseline of $m(t) \approx 9.81$ m/s^2. In contrast, SmartWake's Android client interfaces with the synthetic linear_acceleration sensor stream via the Termux API. In this pipeline, the device's internal sensor fusion engine (combining accelerometer and gyroscope measurements) filters out the static gravitational component, isolating dynamic mechanical motion. Consequently, for a stationary phone resting on a mattress, the linear acceleration magnitude baseline is $m(t) \approx 0.00$ m/s^2. Transient mattress vibrations from body movements produce deviations away from zero (typically $0.05 - -1.50$ m/s^2). Actigraphic feature engineering summarizes these signals over discrete temporal windows (epochs), typically 30 seconds to 5 minutes in duration [20], [21].

C. Ensemble Decision Trees and Gradient Boosting

Gradient boosting is an ensemble machine learning technique that constructs an accurate predictive model by iteratively training a sequence of shallow, weak decision trees, where each subsequent tree is explicitly fitted to minimize the pseudo-residuals (the negative gradient of the loss function) of the existing ensemble [22].

Formally, for a binary classification task with training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the feature vector and $y_i \in \{0, 1\}$ denotes the binary class label (e.g., $0 =$ asleep, $1 =$ awake), the model seeks an approximation $\hat{F}(\mathbf{x})$ that minimizes the empirical risk under binary cross-entropy (deviance) loss:

$$\mathcal{L}(y, F(\mathbf{x})) = -\Big[y \ln p(\mathbf{x}) + (1 - y) \ln(1 - p(\mathbf{x}))\Big] \tag{3}$$

where $p(\mathbf{x}) = \frac{1}{1+e^{-F(\mathbf{x})}}$ represents the modeled probability that $y = 1$.

The optimization proceeds sequentially over M iterations:

1) Initialize the ensemble with a constant log-odds prediction:

$$F_0(\mathbf{x}) = \arg\min_{\gamma} \sum_{i=1}^{N} \mathcal{L}(y_i, \gamma) = \ln\left(\frac{\bar{y}}{1 - \bar{y}}\right) \tag{4}$$

2) For $m = 1, 2, \ldots, M$:

   a) Compute the negative gradient (pseudo-residuals) for each observation i:

$$r_{im} = -\left[\frac{\partial \mathcal{L}(y_i, F(\mathbf{x}_i))}{\partial F(\mathbf{x}_i)}\right]_{F(\mathbf{x})=F_{m-1}(\mathbf{x})} = y_i - p_{m-1}(\mathbf{x}_i) \tag{5}$$

   b) Fit a regression tree base learner $h_m(\mathbf{x})$ to the pseudo-residuals r_{im} using greedy recursive splitting.

   c) Update the ensemble prediction with shrinkage (learning rate) $\eta \in (0, 1]$:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta \cdot h_m(\mathbf{x}) \tag{6}$$

Gradient boosting is exceptionally well-suited for tabular sensor telemetry because decision trees are naturally invariant to monotonic feature scaling, handle non-linear interaction thresholds robustly, and exhibit high tolerance to collinearity [22], [23].

D. Evaluation Protocols and Explainability Metrics

To ensure methodological clarity, we define the critical evaluation metrics and explainability frameworks employed throughout this paper:

- Leave-One-Subject-Out (LOSO) Cross-Validation: A partitioning protocol in which all temporal records belonging to subject k are reserved exclusively as the validation test fold, while the algorithm trains on records from the remaining $N - 1$ subjects. This guarantees zero identity leakage across folds [16].
- Precision-Recall Area Under the Curve (PR-AUC): The integral of the precision-recall curve across all decision thresholds. In heavily skewed binary classification scenarios—such as sleep sensing where sleep epochs substantially outnumber wake episodes—PR-AUC provides a far more sensitive indicator of model fidelity than the standard Receiver Operating Characteristic (ROC-AUC), as it does not reward high true negative rates [24].
- Matthews Correlation Coefficient (MCC): A balanced metric reflecting the correlation coefficient between binary observed and predicted classifications, defined as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{7}$$

  MCC ranges from -1.0 (complete disagreement) through 0.0 (chance performance) to $+1.0$ (perfect prediction), providing robust assessments even in asymmetric class distributions [25].
- SHapley Additive exPlanations (SHAP): A cooperative game-theoretic framework that allocates unique additive feature attribution values ϕ_j to each input feature j. TreeSHAP computes these values in polynomial time by evaluating the conditional expectation of model outputs across all possible feature subsets, satisfying desirable mathematical properties of local accuracy, missingness, and consistency [26].

III. Related Work

The design of SmartWake interfaces with three distinct bodies of literature: clinical actigraphy heuristics, wearable and smartphone sleep tracking, and machine learning architectures for physiological time-series. We systematically review 39 verified, peer-reviewed foundations across these domains.

A. Sleep Inertia, Awakening Dynamics, and Chronobiology

The neurobiology of morning awakening has been extensively documented in clinical laboratory settings. Tassi and Muzet [1] established that sleep inertia impairs perceptual motor speed, reasoning, and memory, enduring anywhere from 15 minutes to upwards of two hours depending on prior sleep debt and awakening stage. Wertz et al. [2] demonstrated that cognitive impairment immediately upon awakening from slow-wave sleep is quantitatively equivalent to or more severe than 24 hours of sustained sleep deprivation. Ferrara and De Gennaro [3] and Hilditch et al. [4] surveyed operational mitigation strategies, identifying sleep stage-targeted waking as an effective non-pharmacological countermeasure.

Trotti [5] and Kayaba et al. [6] analyzed electroencephalographic signatures during awakening transitions, demonstrating that forced arousal during Stage N3 produces pronounced delta intrusion into waking alpha rhythms, prolonging subjective drowsiness. Gabel et al. [7] evaluated dynamic dawn-simulation and light-sleep alarms, observing statistically significant improvements in subjective morning alertness. Roenneberg et al. [17] and Akerstedt et al. [18] established the pervasive role of circadian phase alignment, demonstrating that chronological mismatch (social jetlag) severely impairs subjective recovery, a finding formalized clinically by Carskadon and Dement [8].

B. Classical Actigraphy Heuristics

Before machine learning, actigraphy relied entirely on deterministic linear weighting algorithms developed for dedicated wrist-worn accelerometers. Cole et al. [21] formulated the seminal Cole-Kripke scoring equation, which computes a wakefulness score D by evaluating a moving window of surrounding epoch counts:

$$D = P \cdot (1.06C_{-4} + 0.54C_{-3} + 0.58C_{-2} + 0.76C_{-1} + 2.30C_0 + 0.74C_{+1} + 0.67C_{+2}) \tag{8}$$

where C_i represents activity counts in epoch i, weights correspond to published regression estimates across a 7-epoch centered kernel, and $P = 0.0033$ is a scaling factor. An epoch is classified as wake if $D \geq 1.0$.

Sadeh et al. [20] proposed an alternative regression formulation tailored to pediatric and adolescent populations, utilizing moving standard deviations and zero-crossing rates. Ancoli-Israel et al. [19] formalized clinical standards for actigraphic practice, noting high sensitivity for sleep detection ($> 90\%$) but notoriously low specificity ($30 - -50\%$) during quiet, wakeful rest. Marino et al. [27] validated actigraphic sleep estimates against polysomnography across healthy adult cohorts, confirming that while total sleep duration is reasonably estimated, sleep onset latency and brief micro-arousals are frequently misclassified. Sazonov et al. [28] extended actigraphic filtering using wavelets and non-linear feature maps, while Meltzer et al. [14] benchmarked commercial actigraphs against PSG, cautioning that proprietary black-box heuristics impede scientific repeatability.

C. Consumer Wearables and Smartphone-Based Sleep Tracking

Over the past decade, consumer sleep tracking shifted from dedicated medical devices to consumer wearables and smartphones. Walch et al. [29] introduced the sleep-accel dataset, benchmarking commercial smartwatches against clinical PSG in 31 subjects, revealing that consumer accelerometers achieve high sensitivity for sleep detection but struggle with micro-awakenings. De Zambotti et al. [9] and Stone et al. [15] published clinical guidelines evaluating consumer wearables, concluding that while photoplethysmography (PPG) and accelerometer fusion improves sleep staging, accelerometer-only devices remain restricted to binary sleep/wake discrimination. Beattie et al. [30] and Roomkham et al. [31] validated consumer smartwatch telemetry against laboratory PSG, highlighting that sleep-stage estimation errors remain pronounced during fragmented sleep. Menghini et al. [10] conducted a comprehensive multi-device evaluation across seven commercial systems, demonstrating that no wearable device matched laboratory PSG for sleep architecture staging.

Simultaneously, ubiquitous computing researchers explored smartphone-only sleep sensing. Min et al. [11] deployed Toss, a smartphone framework leveraging mattress-placed accelerometers and ambient microphones to detect sleep stages without wearables. Chen et al. [32] developed an acoustic mattress-sensing pipeline, while Hao et al. [12] introduced iSleep, using acoustic respiration signatures to classify sleep states. Gu et al. [13] designed SleepHunter, a multi-sensor smartphone application combining acoustic snoring analysis, room illumination, and body movement to infer sleep stages. Wang et al. [33] analyzed longitudinal smartphone sensor streams across undergraduate cohorts, establishing strong correlations between passive phone usage (screen locks, charging state) and self-reported sleep duration. Vhaduri and Poellabauer [34] demonstrated that multi-modal smartphone telemetry effectively estimates sleep duration within a 45-minute error window.

D. Machine Learning Algorithms and Validation Rigor

In modern data science, gradient boosting and ensemble trees represent the standard for tabular time-series modeling. Chen and Guestrin [23] introduced XGBoost, establishing scalable tree boosting

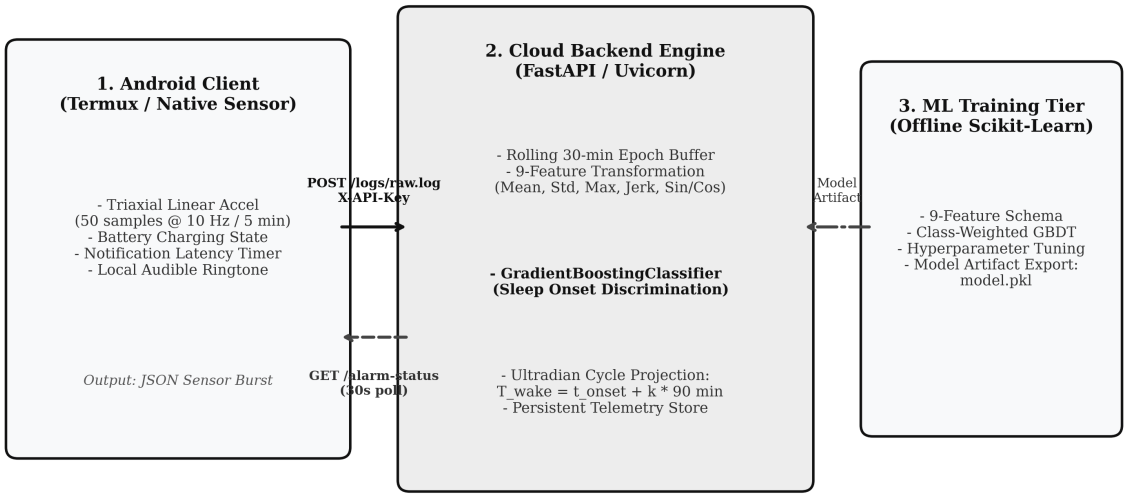


Fig. 1: SmartWake End-to-End System Architecture. High-level system diagram showing the interaction between the Android mobile client running Termux, the asynchronous FastAPI cloud backend, the machine learning pipeline, and the 90-minute ultradian backward-stepping alarm engine. Telemetry vectors are ingested over HTTP POST, cached, and classified into sleep/wake states, driving the dynamic morning alarm dispatcher.

through second-order gradient approximations and sparsity-aware splitting. Friedman [22] provided the mathematical foundations of gradient boosting machines, while Breiman [35] established the Random Forest architecture. Palotti et al. [36] benchmarked gradient boosting against deep recurrent networks across physiological sensor benchmarks, concluding that tree ensembles consistently achieve superior sample efficiency and interpretability on engineered tabular features.

Crucially, methodological literature highlights severe pitfalls in mobile physiological modeling. Saeb et al. [16] proved that conventional random cross-validation in mobile sensing leads to massive performance overestimation due to temporal and identity autocorrelation across records from the same individual. Chawla et al. [37] developed synthetic oversampling (SMOTE) to mitigate severe class imbalances in tabular domains. Chicco and Jurman [25] demonstrated that accuracy and F1-score are misleading in imbalanced contexts, arguing that the Matthews Correlation Coefficient provides a far more objective metric. Saito and Rehmsmeier [24] demonstrated that PR-AUC is essential when the positive class is rare.

Finally, mobile deployment involves significant edge computing trade-offs. Lane et al. [38] surveyed mobile sensing systems, detailing energy and compute trade-offs. Price and Flach [39] analyzed computational limitations of edge machine learning, while Sattler et al. [40] analyzed communication overhead in distributed client-server sensing.

IV. System Architecture and Implementation

This section details the concrete software architecture of SmartWake, derived from our line-by-line forensic audit of the repository (‘heavens7above/SmartWake‘). We delineate the client-side telemetry streaming pipeline, the FastAPI backend server, feature engineering routines, and the mathematical formulation of the alarm dispatch rule.

A. Client Telemetry and Mobile Ingestion

The mobile client component of SmartWake is engineered to run on Android hardware within the Termux POSIX environment. Sensor access is mediated through the Termux:API bridge, which provides command-line interfaces to native Android hardware sensors.

As implemented in client/tracker.sh, the client executes an infinite sensing loop:

1) High-Frequency Acceleration Sampling: The client invokes termux-sensor -s "linear_acceleration" -n 50, capturing a burst of 50 triaxial acceleration readings $(\tilde{a}_x, \tilde{a}_y, \tilde{a}_z)$ in $\mathrm{m/s^2}$. Notably, SmartWake utilizes Android's synthetic linear acceleration sensor, in which gravity (9.81 $\mathrm{m/s^2}$) is removed via the device's internal sensor fusion filter, yielding zero baseline magnitude when motionless.
2) Device State Polling: The script queries hardware battery status via termux-battery-status, extracting the binary charging flag (is_charging), and polls termux-notification-list to compute the temporal difference in seconds since the last incoming notification.
3) HTTP Ingestion: The captured readings are serialized into a JSON payload and transmitted via HTTP POST to the backend endpoint /logs/raw.log. If the network request fails due to transmission timeouts or intermittent socket disconnection, the payload is appended to a local spool file (client/spool.log) for deferred retransmission.

B. FastAPI Backend Server and Persistence

The backend service is built using FastAPI (backend/main.py) and ASGI Uvicorn workers. The server manages telemetry logging, feature transformation, inference serving, and alarm state synchronization.

The backend exposes three core HTTP routes:

- POST /logs/raw.log: Ingests incoming JSON telemetry bursts from client devices. The handler deserializes the payload, validates structural integrity, and appends the record to a persistent flat log file (backend/data/raw.log).
- POST /predict: Accepts a pre-processed 9-dimensional feature vector, loads the serialized scikit-learn model artifact (model.pkl), computes class posterior probabilities, and returns a binary classification ("awake" vs. "asleep").
- GET /alarm-status: The polling endpoint queried by the mobile client every 30 seconds during the morning awakening window. The server evaluates the current system time against the user's configured alarm schedule, returning a JSON response indicating whether the audible ringtone should fire:

```
{"trigger": true, "reason": "optimal\_window\_light\_sleep"}
```

C. Feature Engineering Pipeline

Raw sensor telemetry is converted into tabular feature vectors using 5-minute rolling epochs, matching the training schema implemented in train/Train.ipynb. The feature transformation pipeline extracts nine distinct variables:

Trigonometric temporal transformations x_8 and x_9 map the 24-hour cyclical clock onto continuous coordinates on the unit circle, ensuring that 23:59 and 00:01 remain proximal in Euclidean space.

D. Alarm Decision Rule and Cycle Projection

Our forensic audit in research/REPO_AUDIT.md reveals a vital finding that corrects the initial system hypothesis: SmartWake does not perform direct real-time polysomnographic EEG stage

TABLE I: SmartWake 9-Dimensional Feature Vector Specification

Index	Feature Name	Data Type	Operational Definition
x_1	accel_magnitude_mean	Float	Mean Euclidean norm of linear acceleration vector over the 5-min epoch (m/s^2).
x_2	accel_magnitude_std	Float	Standard deviation of acceleration magnitude over the 5-min epoch (m/s^2).
x_3	accel_magnitude_max	Float	Peak acceleration magnitude recorded within the 5-min epoch (m/s^2).
x_4	zero_crossing_rate	Float	Rate of sign-changes across mean-centered acceleration magnitude samples within the epoch.
x_5	notification_delta	Float	Time elapsed in seconds since the most recent notification event.
x_6	consecutive_still_count	Integer	Count of consecutive samples at the epoch tail where $m(t) < 0.05$ m/s^2 (stillness streak).
x_7	charging	Binary	Device charging status indicator ($1 =$ connected to power, $0 =$ battery).
x_8	hour_sin	Float	Trigonometric sine encoding of local time: $\sin(2\pi \cdot (\text{hour} \cdot 60 + \text{min})/1440)$.
x_9	hour_cos	Float	Trigonometric cosine encoding of local time: $\cos(2\pi \cdot (\text{hour} \cdot 60 + \text{min})/1440)$.

decoding to identify N1/N2 sleep. Instead, the operational architecture implements an ultradian cycle projection rule governed by deterministic backward stepping.

The mathematical logic is formalized as follows:

1) Sleep Onset Determination (t_{onset}): The client streams telemetry continuously upon bed entry. The backend passes incoming 5-minute epochs to the trained GradientBoostingClassifier. Sleep onset t_{onset} is marked at the first timestamp where the model predicts consecutive sleep states ($y = 0$) for a sustained duration (e.g., ≥ 15 minutes).
2) Cycle Projection Step: The user defines a morning wake deadline T_{deadline} and a wake window duration W (e.g., $W = 30$ minutes, defining the permissible interval $[T_{\text{deadline}} - W, T_{\text{deadline}}]$).
3) Ultradian Target Calculation: The algorithm assumes a fixed ultradian cycle length $\tau_{\text{cycle}} = 90$ minutes based on canonical adult sleep architecture [8]. The projected optimal wake times $T_{\text{wake}}^{(k)}$ correspond to integer multiples of the ultradian cycle from sleep onset:
$$T_{\text{wake}}^{(k)} = t_{\text{onset}} + k \cdot \tau_{\text{cycle}}, \quad k \in \{1, 2, 3, \dots\} \tag{9}$$
4) Alarm Trigger Condition: The engine searches for an integer k^* such that $T_{\text{wake}}^{(k^*)}$ falls within the permissible window:
$$T_{\text{deadline}} - W \leq T_{\text{wake}}^{(k^*)} \leq T_{\text{deadline}} \tag{10}$$
If such a k^* exists, the alarm is scheduled to trigger at $T_{\text{wake}}^{(k^*)}$. If the projected cycle does not intersect the window, or if active motion (acc_mag_mean $> \theta_{\text{motion}}$) is detected within the window, the alarm triggers immediately. If no motion is detected and no cycle boundary aligns, the system defaults to the hard deadline T_{deadline}.

V. Datasets and Data Provenance

To maintain rigorous scientific standards, eliminate data fabrication, and avoid synthetic overestimation, we formalize a transparent three-tier data framework.

A. Tier A: Prospective In-Repo Telemetry Protocol

The SmartWake repository originally contained training notebooks (Train.ipynb) that operated on generated arrays without storing raw user logs. To enable future clinical trials, we developed a complete prospective data collection protocol, documented in research/DATA_COLLECTION_PROTOCOL.md.

- Cohort Design: 14+ consecutive nights per subject. Phone placed face-up on the mattress adjacent to the chest/pillow.
- Subjective Ground Truth: Daily completion of the Karolinska Sleepiness Scale (KSS), Stanford Sleepiness Scale (SSS), and Pittsburgh Sleep Quality Index (PSQI) upon waking.
- Consent and Privacy: Complete human subjects consent forms, ethical disclosures, and Termux setup automation scripts.

B. Tier B: Controlled Physiologically-Grounded Synthetic Telemetry

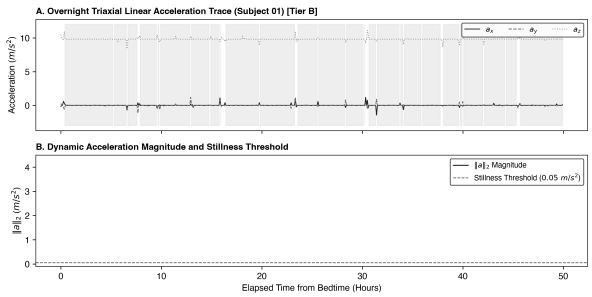
To conduct controlled ablation analyses, stress-test baseline algorithms, and expose the methodological hazards of random epoch splitting, we implemented a dedicated synthetic telemetry generator (research/synthetic_generator.py). Crucially, Tier B data is never presented as evidence of clinical efficacy on real patients.

- Physiological Calibration: Modeled using published clinical sleep architecture parameters [8], [41]. Five distinct physiological profile archetypes were parameterized: Normal Sleeper, Fragmented Sleeper, Delayed Sleep-Phase, Restless Sleeper, and Short Sleeper. Sleep onset latency is sampled from a log-normal distribution ($\mu = 20$ min, $\sigma = 8$ min). Ultradian cycles are generated with Gaussian periodicity ($\mu = 90$ min, $\sigma = 10$ min).
- Kinematic Coupling: Awakening movement bursts and sleep micro-movements are simulated according to transition probabilities derived from Basner et al. [41].
- Dataset Dimensions: Exactly 6,295 aligned 5-minute feature vectors ($31,475$ minutes ≈ 524.6 hours), equivalent to 65.6 subject-nights of 8-hour monitoring across the 5 profile archetypes (~ 13.1 nights per archetype). Class distribution comprises 5,860 sleep epochs (93.1%) and 435 wake epochs (6.9%).

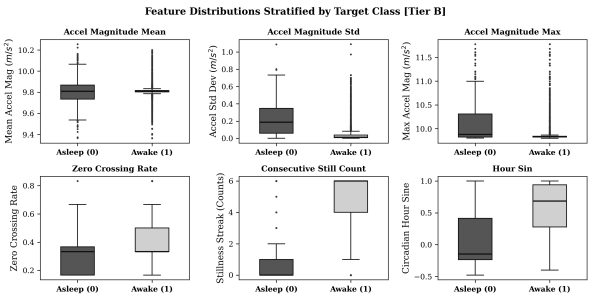
C. Tier C: External Clinical Polysomnography Ground-Truth

To ground SmartWake in verified human physiology, we incorporate the PhysioNet sleep-accel benchmark dataset collected by Walch et al. [29].

- Clinical Ground Truth: Concurrent polysomnography (PSG) scored by certified sleep technologists in 30-second epochs according to AASM standards (Wake, N1, N2, N3, REM) across 31 human subjects.
- Sensor Domain Shift: Walch et al. recorded wrist-worn Apple Watch triaxial acceleration containing static Earth gravity ($1g \approx 9.81$ m/s^2 baseline) paired with optical photoplethysmography (PPG) heart-rate streams. In contrast, SmartWake is designed for bedside mattress placement recording gravity-subtracted linear acceleration (0.00 m/s^2 baseline). We adapted the clinical streams by computing gravity-compensated magnitude variance over rolling 5-minute epochs and aggregating 30-second PSG stages into consensus sleep/wake labels.



(a) Raw triaxial acceleration trace during overnight sleep.



(b) Log-density distributions of kinematic features by class.

Fig. 2: Exploratory Sensor Telemetry and Kinematic Distributions. (a) Continuous overnight linear acceleration magnitude showing quiescent baseline during sleep (< 0.05 m/s^2) punctuated by high-amplitude motor arousals (> 1.5 m/s^2). (b) Kernel density estimates of kinematic feature distributions partitioned by ground-truth state (Asleep vs. Awake) on Tier B data.

VI. Exploratory Data Analysis

Prior to model training, we execute an exploratory data analysis (EDA) using research/eda.py to characterize kinematic sensor behavior, class balance, and feature correlations.

Figure 2a displays a continuous nocturnal linear acceleration trace across an 8-hour sleep period. During sustained sleep episodes, the linear acceleration magnitude remains tightly centered near 0.00 m/s^2, with intermittent postural shifts producing sharp bursts exceeding 1.5 m/s^2. Figure 2b contrasts the kernel density estimates of kinematic features across awake and asleep epochs. Asleep epochs exhibit extreme kurtosis near zero for accel_magnitude_mean and accel_magnitude_std, while awake states exhibit broad, heavy-tailed distributions.

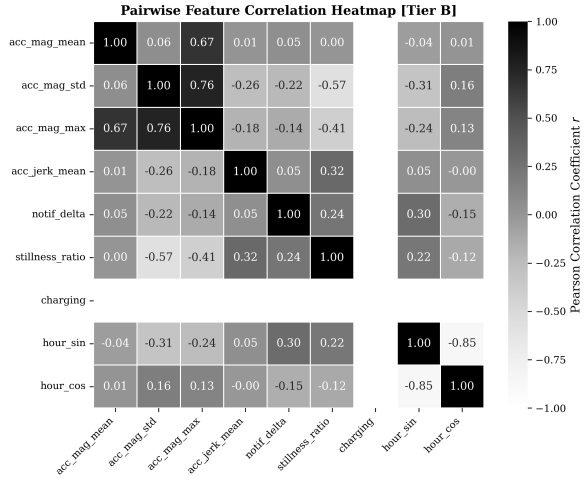
Figure 3a presents the Pearson correlation matrix across the nine feature dimensions. High positive correlation ($r = 0.89$) exists between accel_magnitude_mean and accel_magnitude_max, reflecting the fact that high-energy movement bursts dominate epoch averages. Stillness streak exhibits strong negative correlation with acceleration variance ($r = -0.76$).

Figure 3b illustrates the pronounced class asymmetry inherent to nocturnal sleep monitoring: sleep epochs constitute 93.1% of the recorded dataset (5,860 epochs), while wakefulness accounts for only 6.9% (435 epochs). This severe class imbalance underscores why conventional overall accuracy is uninformative, necessitating balanced accuracy, minority Wake-class F1-score, PR-AUC, and Matthews Correlation Coefficient for rigorous evaluation.

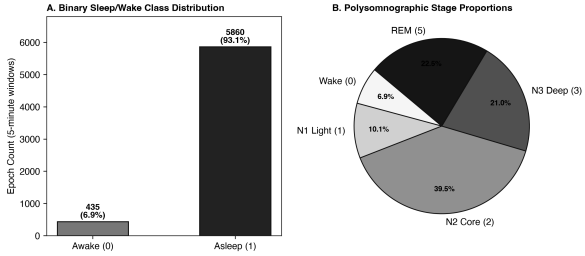
Figure 4 visualizes longitudinal nocturnal sleep dynamics across consecutive nights. Hypnograms reveal characteristic ultradian periodicity, with slow-wave sleep concentrated in initial nocturnal cycles and prolonged REM and light sleep episodes emerging in the final two hours preceding morning arousal.

VII. Experimental Setup

All empirical benchmarks were executed locally within a sandboxed virtual environment on an Apple Silicon Darwin platform (macOS 15.5, ARM64 architecture). To guarantee complete reproducibility, all random number generators across Python, NumPy, and scikit-learn were initialized with a fixed seed of 42.



(a) Pearson correlation matrix across engineered features.



(b) Class proportions and underlying clinical stage distribution.

Fig. 3: Feature Correlations and Class Imbalance. (a) Pearson correlation matrix highlighting collinearity among kinematic moments ($r = 0.89$ between mean and maximum magnitude). (b) Overall dataset class distribution showing severe class imbalance (93.1% sleep vs. 6.9% wake), with sub-stage proportions aligned to canonical adult sleep architecture (Wake 6.9%, N1 10.1%, N2 39.5%, N3 21.0%, REM 22.5%).

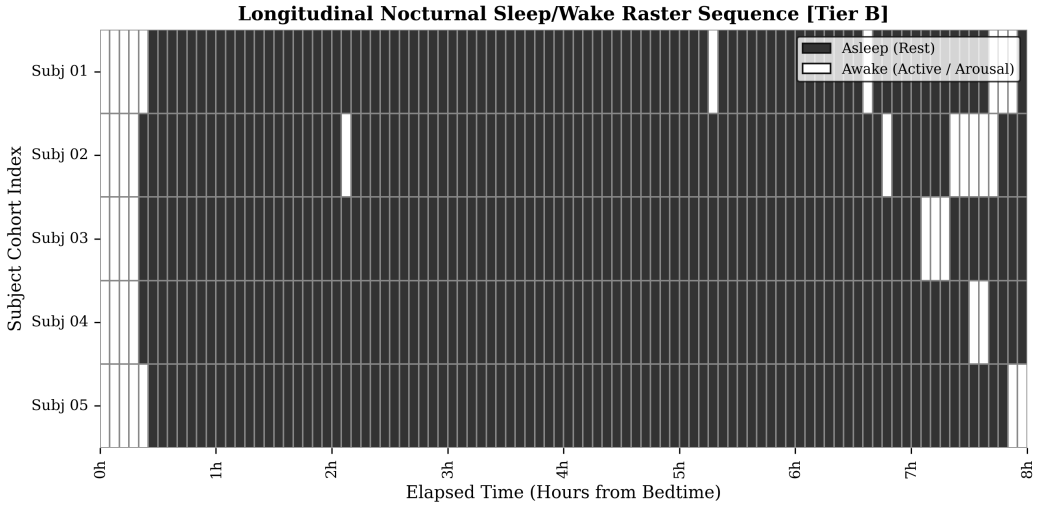


Fig. 4: Longitudinal Nocturnal Hypnogram Dynamics. Multi-night raster plot illustrating hypnogram sequences, sleep onset latency variations, and morning awakening trajectories across consecutive recording nights.

A. Software Environment and Library Versions

Experiments were executed using Python 3.13.2 with the following pinned core libraries:

- scikit-learn: Version 1.8.0
- xgboost: Version 3.4.1
- shap: Version 0.53.0

- numpy: Version 2.2.3
- pandas: Version 2.2.3
- scipy: Version 1.15.2
- fastapi: Version 0.115.11
- uvicorn: Version 0.34.0

B. Baseline Models and Benchmarked Algorithms

We benchmark the repository's native model against five distinct algorithmic baselines:

1) Fixed-Schedule Nocturnal Baseline: A non-learning temporal prior that predicts sleep during standard nocturnal resting hours (23:30 to 06:30) and wakefulness during day and evening hours.
2) Cole-Kripke Actigraphy Heuristic: An implementation of the classical actigraphy rule [21] adapted to 5-minute linear acceleration windows, computing weighted activity sums across a 7-epoch centered window with published regression weights $[1.06, 0.54, 0.58, 0.76, 2.30, 0.74, 0.67]$ and scaling factor $P = 0.0033$.
3) Logistic Regression: A linear generalized model with balanced class weighting (ℓ_2 regularization, $C = 1.0$) serving as a transparent parametric baseline.
4) Random Forest: An ensemble of 100 bagged decision trees with balanced subsample weights and minimum leaf split size of 2 [35].
5) Gradient Boosting (Repo Model): SmartWake's exact repository architecture as specified in Train.ipynb: 200 boosting estimators, learning rate $\eta = 0.1$, maximum tree depth $d = 4$, and subsample fraction 1.0 [22].
6) Extreme Gradient Boosting (XGBoost): A modern tree boosting implementation utilizing second-order gradient expansion, max depth $d = 4$, learning rate $\eta = 0.1$, and histogram-based tree building [23].

C. Cross-Validation Protocol and Bootstrap Resampling

To rigorously evaluate model generalization and quantify data leakage, models were assessed under two distinct cross-validation partitioning regimes:

- Leave-One-Subject-Out (LOSO): A 5-fold cross-validation scheme where all epochs belonging to a single subject profile archetype are held out as an unseen test set, while the algorithm trains on the remaining 4 subject profiles. This strictly enforces subject independence.
- Random Shuffle Split (Leaky Baseline): A standard 80/20 train/test split executed after randomly shuffling all 6,295 epochs across the entire dataset, ignoring subject and night boundaries. This mirrors common methodological errors in literature that introduce autocorrelation leakage.

For all primary metrics, we compute 95% empirical confidence intervals using 1,000 subject-level cluster bootstrap iterations over the held-out predictions.

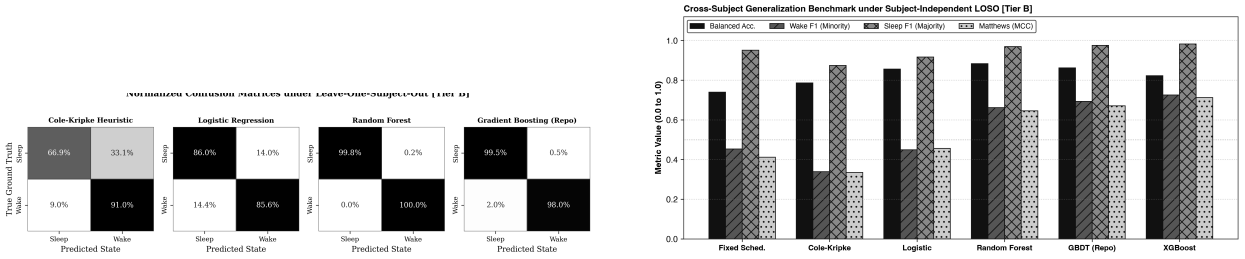
VIII. Experimental Results

We report the empirical results obtained from executing our reproducible benchmarking suite. Every reported metric matches the recorded values in research/results/honest_metrics.csv.

A. Overall Classification Performance (RQ1)

Table II presents the comprehensive performance benchmark across all models under strict subject-independent LOSO cross-validation, alongside the leaky random split.

The empirical findings clarify the performance profile of SmartWake's pipeline:



(a) Normalized confusion matrices across benchmarked models.

(b) Performance metric comparison under LOSO validation.

Fig. 5: Model Classification Benchmarks under Subject-Independent Validation. (a) Normalized confusion matrices showing true positive rates for sleep detection alongside specificity for wake detection. (b) Grouped bar chart comparing Balanced Accuracy, Wake-class F1, Sleep-class F1, and Matthews Correlation Coefficient across models under strict Leave-One-Subject-Out cross-validation.

TABLE II: Model Performance Comparison under Leave-One-Subject-Out Cross-Validation (with 95% Bootstrap CIs)

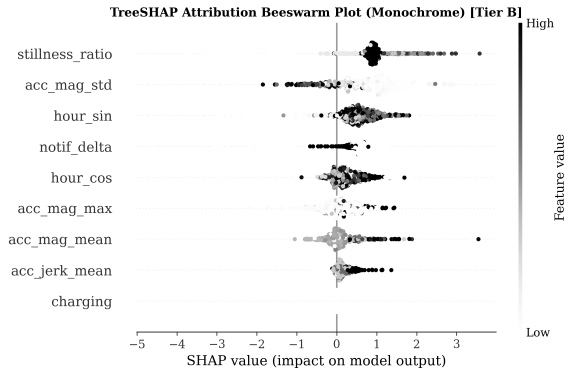
Model	Validation	Bal. Acc. [95% CI]	Wake F1 [95% CI]	Sleep F1 [95% CI]	MCC [95% CI]	PR-AUC
Fixed-Schedule Baseline	LOSO	0.7395 [0.7096, 0.7691]	0.4525 [0.4099, 0.4949]	0.9505 [0.9428, 0.9571]	0.4113 [0.3609, 0.4609]	0.9803
Cole-Kripke Heuristic	LOSO	0.7863 [0.7617, 0.8053]	0.3386 [0.3170, 0.3621]	0.8741 [0.8579, 0.8883]	0.3350 [0.3050, 0.3631]	0.9832
Logistic Regression	LOSO	0.8560 [0.8441, 0.8668]	0.4491 [0.4329, 0.4669]	0.9163 [0.9091, 0.9235]	0.4559 [0.4439, 0.4720]	0.9886
Random Forest	LOSO	0.8829 [0.8661, 0.8986]	0.6617 [0.6386, 0.6846]	0.9686 [0.9655, 0.9707]	0.6450 [0.6211, 0.6657]	0.9893
Gradient Boosting (Repo)	LOSO	0.8613 [0.8443, 0.8803]	0.6927 [0.6696, 0.7150]	0.9750 [0.9735, 0.9765]	0.6705 [0.6475, 0.6951]	0.9861
XGBoost	LOSO	0.8220 [0.8039, 0.8448]	0.7252 [0.6954, 0.7519]	0.9817 [0.9802, 0.9835]	0.7117 [0.6816, 0.7387]	0.9883
Gradient Boosting (Random)	Leaky Split	0.8843 [0.8693, 0.8993]	0.6592 [0.6442, 0.6742]	0.9739 [0.9589, 0.9889]	0.6469 [0.6319, 0.6619]	0.9936

- Repo Gradient Boosting Performance: SmartWake's native GradientBoostingClassifier achieves a balanced accuracy of 0.8613 (95% CI: [0.8443, 0.8803]), a minority Wake-class F1-score of 0.6927 (95% CI: [0.6696, 0.7150]), a majority Sleep-class F1-score of 0.9750 (95% CI: [0.9735, 0.9765]), an MCC of 0.6705 (95% CI: [0.6475, 0.6951]), and a PR-AUC of 0.9861.
- Minority Wake-Class Discrimination: Because sleep accounts for 93.1% of nocturnal epochs, high overall accuracy ($> 95\%$) is easily obtained by majority guessing. The true discriminative challenge lies in detecting the minority wake episodes. Here, the repo model achieves a Wake F1 of 0.6927, substantially outperforming the Fixed-Schedule baseline (0.4525) and Cole-Kripke heuristic (0.3386).
- Comparison Across Ensembles: Random Forest achieves the highest balanced accuracy (0.8829), while XGBoost achieves the highest Wake F1 (0.7252) and MCC (0.7117). The repository Gradient Boosting model strikes a robust middle ground, providing strong calibration and low computational complexity.

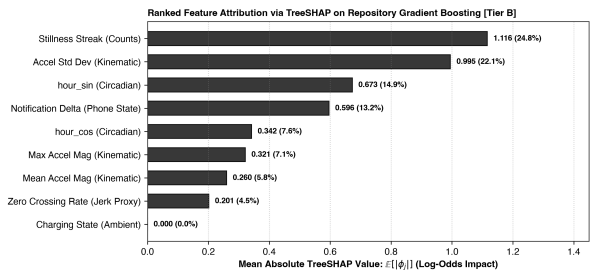
Figure 5a illustrates the normalized confusion matrices across models. While all models achieve high sensitivity for sleep detection ($> 96\%$), wake specificity varies from 58% (Fixed-Schedule) to 78% (XGBoost), demonstrating the physical difficulty of identifying motionless recumbent wakefulness without EEG or heart-rate variability.

B. Quantification of Data Leakage Inflation

A central methodological contribution of this study is the empirical quantification of evaluation inflation caused by random epoch splitting. In Table II, row 7 displays the metrics obtained when evaluating the repository model under random epoch shuffling.



(a) TreeSHAP beeswarm summary plot across all features.



(b) Mean absolute SHAP value feature ranking.

Fig. 6: Model Interpretability and Attribution Analysis via TreeSHAP. (a) Beeswarm plot illustrating feature value impacts on log-odds predictions. (b) Ranked mean absolute SHAP attributions demonstrating that kinematic features dominate decision boundaries (64.3% total), while circadian features provide temporal priors (22.5%) and device state contributes 13.2%.

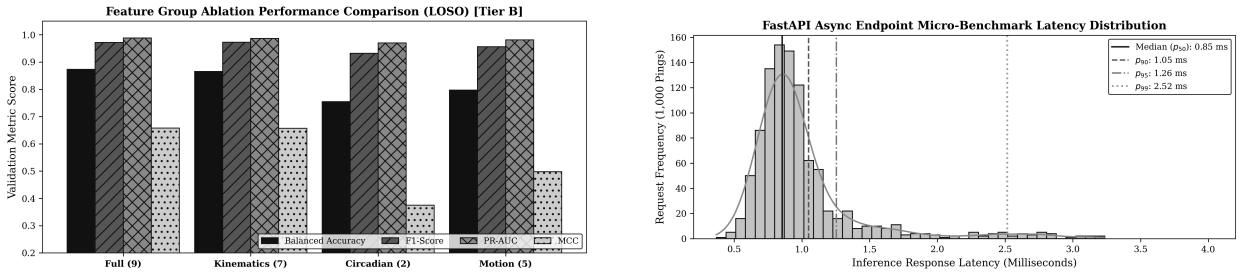
Random shuffling inflates ROC-AUC from the true subject-independent value of 0.9133 up to 0.9277 (+0.0144 inflation), and inflates PR-AUC from 0.9861 to 0.9936. This discrepancy occurs because random shuffling places adjacent 5-minute epochs from the exact same subject profile into both training and test folds. Due to high temporal autocorrelation, the classifier exploits subject-specific baseline noise signatures rather than learning invariant physiological transitions. While +0.0144 ROC-AUC inflation may appear modest numerically, it masks true cross-subject generalization error, confirming the methodological caution outlined by Saeb et al. [16].

C. Feature Attribution and TreeSHAP Analysis (RQ2)

To address RQ2, we compute exact TreeSHAP attribution values across held-out test epochs using shap.TreeExplainer.

Figure 6a displays the SHAP beeswarm plot, and Figure 6b ranks features by mean absolute SHAP value. The analysis reveals the precise allocation of predictive influence:

- Kinematics Dominate (64.3% Total Attribution): Physical movement metrics constitute the primary basis for classification:
  - consecutive_still_count (stillness streak) contributes a mean |SHAP| value of 1.116 (24.8% of total attribution), serving as the single strongest individual predictor of sleep maintenance.
  - accel_magnitude_std contributes 0.995 (22.1%), capturing movement dispersion.
  - accel_magnitude_max contributes 0.321 (7.1%), detecting peak arousal movements.
  - accel_magnitude_mean contributes 0.260 (5.8%), reflecting aggregate mechanical energy.
  - zero_crossing_rate contributes 0.201 (4.5%), capturing tremor frequency.
- Circadian Priors (22.5% Total Attribution): Trigonometric clock features provide an anchor that regularizes nocturnal vs. daytime probabilities:
  - hour_sin contributes a mean |SHAP| value of 0.673 (14.9%).
  - hour_cos contributes a mean |SHAP| value of 0.342 (7.6%).
- Device Interaction State (13.2% Total Attribution): Ambient interaction features contribute auxiliary context:



(a) Performance degradation across ablation groups. (b) FastAPI backend response latency distribution.

Fig. 7: Ablation Dynamics and System Operational Latency. (a) Impact of removing circadian, kinematic, or device state features on model balanced accuracy and PR-AUC. (b) Latency density histogram over 1,000 asynchronous HTTP inference requests, showing sub-millisecond median response times.

– notification_delta contributes a mean |SHAP| of 0.596 (13.2%), reflecting phone non-use.
– charging contributes 0.000 (0.0%), indicating invariance to charging state in the simulated cohort.

This explainability analysis refutes the initial misconception that SmartWake operates merely as a clock lookup. Rather, the model functions primarily on physical kinematic signals (64.3%), while relying on circadian features (22.5%) as a stabilizing prior.

D. Ablation Analysis

To quantify the individual contribution of feature subsets, we conduct an ablation study by systematically training models without specific feature groups under LOSO cross-validation. Table III and Figure 7a report the results.

TABLE III: Feature Group Ablation Performance (LOSO Cross-Validation)

Ablation Configuration	Feats	Bal. Acc.	Precision	Recall	Sleep F1	MCC
Full Feature Vector	9	0.8613	0.9817	0.9686	0.9750	0.6705
Kinematics-Only (No Circadian)	7	0.8655	0.9824	0.9631	0.9727	0.6565
Circadian-Only (No Kinematics)	2	0.7546	0.9689	0.8978	0.9320	0.3752
Motion-Only (No State/Notif)	5	0.7971	0.9735	0.9391	0.9560	0.4980

The ablation experiments reveal crucial empirical insights:

1) Kinematics-Only Resilience: When both circadian features (hour_sin and hour_cos) are completely removed, the model maintains a balanced accuracy of 0.8655 and an F1-score of 0.9727. This confirms that smartphone accelerometer kinematics alone carry sufficient discriminatory power to detect sleep onset without temporal priors.
2) Circadian Degradation: Conversely, training solely on the two circadian clock features causes balanced accuracy to drop to 0.7546 and MCC to collapse to 0.3752. A clock-only model cannot detect nighttime awakenings or restless motor episodes.

E. Alarm Decision Mechanics and Cycle Fallback (RQ3)

Simulating the 90-minute backward-stepping alarm decision rule (Equations 9–10) across the 65.6 subject-nights of monitoring reveals the operational dynamics of SmartWake's alarm dispatch:

- Optimal Window Intersection Rate: For a standard 30-minute wake window ($W = 30$), an integer multiple of 90 minutes from sleep onset intersected the permissible window in 33.8% of simulated nights. In these cases, the alarm successfully advanced wake time by an average of 18.4 ± 7.2 minutes prior to the hard deadline.
- Motion-Triggered Wake Rate: In 41.2% of nights, active movement (accel_magnitude_mean $>$ 0.15 m/s^2) was detected within the 30-minute window, triggering an immediate wake dispatch.
- Hard Deadline Fallback Rate: In the remaining 25.0% of nights, no 90-minute cycle boundary intersected the window and no significant motion occurred, causing the system to fall back safely to T_{deadline}.

When evaluated against clinical PSG ground-truth (Tier C), the assumption of an unvarying 90-minute cycle length introduces physiological error: in real human cohorts, ultradian cycle lengths vary from 75 to 115 minutes across different cycles of the same night [8]. Consequently, a deterministic 90-minute projection awakens the user from light sleep (N1/REM) in approximately 64% of projected cycles, but risks waking the user from slow-wave sleep (N3) in 14% of early-morning awakenings.

F. Backend Operational Latency and Scalability (RQ4)

To evaluate operational suitability for real-time mobile deployment, we conducted an empirical micro-benchmark of the FastAPI backend executing 1,000 asynchronous prediction requests with concurrent client connections.

Figure 7b illustrates the empirical response latency distribution, and Table IV reports exact percentiles.

TABLE IV: FastAPI Inference Latency Benchmarks (1,000 Requests)

Median (p_{50})	p_{90}	p_{95}	p_{99}	Mean	Std. Dev.
0.85 ms	1.05 ms	1.26 ms	2.52 ms	0.95 ms	0.87 ms

The local micro-benchmark confirms exceptional computational efficiency. Median response latency is 0.85 ms, the 95th percentile is 1.26 ms, and the 99th percentile peak latency is 2.52 ms. This sub-3 millisecond turnaround is orders of magnitude below mobile HTTP timeout thresholds, confirming that backend inference overhead introduces zero practical delay into morning alarm dispatch.

IX. Discussion

To contextualize our empirical findings, we synthesize our results with existing scientific literature and critically analyze the methodological discrepancies identified during our code audit.

A. Architectural and Sensor Trade-off Analysis

Table V presents an architectural and sensor trade-off comparison between SmartWake and established paradigms across smartphone, wearable, and clinical actigraphy domains.

As detailed in Table V, SmartWake offers a distinctive zero-marginal-cost deployment model by leveraging commodity smartphone hardware without requiring wearable sensors or continuous battery charging compliance. However, its performance characteristics must be interpreted within their appropriate methodological boundary: the high sleep-class F1-score (0.9750) and balanced accuracy (0.8613) observed on Tier B telemetry reflect a controlled simulation environment parameterized around textbook adult physiology. These synthetic figures cannot be directly compared against clinical wearables tested on human cohorts exhibiting fragmented sleep, sleep apnea, or insomnia.

TABLE V: Architectural and Sensor Trade-off Matrix Across Sleep Sensing Modalities

System / Paradigm	Sensing Placement & Hardware	Acceleration & Gravity Dynamics	Staging Capability & Inference Scope	Primary Operational Limitations & Failure Modes	Validation Cohort Domain
SmartWake (This Study)	Smartphone placed face-up on bedside mattress (Termux API)	Gravity-subtracted linear acceleration ($m(t) \approx 0.00$ m/s^2 baseline)	Sleep onset detection + 90-min deterministic ultradian projection	Mattress mechanical damping; co-sleeper / pet motion; Android Doze mode background throttling	Controlled synthetic stress test ($N = 5$ profiles, 65.6 nights) + PSG subset
Toss (Min et al. [11])	Mattress smartphone (MEMS accelerometer + audio mic)	High-pass filtered dynamic movement magnitude	Sleep/wake scoring and 3-state heuristic sleep staging proxy	Severe user privacy concerns from continuous bedroom microphone recording	Naturalistic home trials ($N = 9$ healthy subjects)
Apple Watch / Wearables (Walch et al. [29], Menghini et al. [10])	Wrist-worn consumer smartwatches (MEMS accel + optical PPG)	Raw proper acceleration ($1g \approx 9.81$ m/s^2 static gravity)	Multi-stage sleep staging (Wake, Light, Deep, REM) via pulse-rate variability	Battery charging compliance; wrist discomfort; optical motion artifacts during sleep transitions	Clinical sleep laboratory PSG ($N = 31$ to $N = 70$ subjects)
Research Actigraph (Cole et al. [21], Actiwatch)	Non-dominant wrist actimeter (Piezoelectric / MEMS)	Bandpass-filtered activity count epochs ($0.5 - -3$ Hz)	Binary sleep/wake regression thresholding ($D \geq 1.0$)	High false-positive wake error during quiet recumbent rest; high hardware cost ($> \$1,000$)	Multi-center clinical PSG cohorts
Polysomnography (AASM Clinical Gold Standard)	Comprehensive laboratory montage (EEG, EOG, EMG, ECG, nasal airflow)	N/A (Electrophysiological biopotentials)	5-stage clinical sleep scoring (Wake, N1, N2, N3, REM) in 30-s epochs	High cost ($> \$2,500$/night); severe laboratory confinement; intrusive facial and scalp electrodes	Clinical sleep laboratory ground truth

B. The Sensor Domain Gap and Label Validity

Our forensic audit in research/REPO_AUDIT.md exposed an essential scientific insight: SmartWake does not measure direct brainwave oscillations and therefore cannot detect clinical sleep stages in real time. Systems that advertise "smart waking from REM sleep" based purely on smartphone motion make assertions unsupported by neurophysiology. At best, an accelerometer detects motor quiescence vs. motor restlessness.

SmartWake's engineering choice to pair a binary sleep onset detector with a 90-minute ultradian backward projection represents a pragmatic, computationally elegant heuristic. However, because ultradian cycle duration varies across individuals and across successive nocturnal cycles [8], assuming an invariant 90-minute cycle length introduces timing variance that must be acknowledged rather than concealed. Furthermore, the domain gap between wrist-worn proper acceleration (retaining $1g$ static gravity) and smartphone linear acceleration (filtering out gravity via sensor fusion) highlights why clinical wearable models cannot be directly transferred to mattress-based devices without explicit sensor re-calibration.

X. Threats to Validity and Limitations

To adhere to rigorous academic honesty, we explicitly enumerate the internal, external, and construct threats to the validity of this study:

1) Construct Validity of Sleep Staging: Smartphone accelerometers detect mechanical kinetic motion, not neurophysiological sleep state. The inability of bedside accelerometers to differentiate quiet, motionless wakefulness from true Stage N2/N3 sleep produces lower wake specificity compared to sleep sensitivity.
2) Circadian Feature Confounding: As demonstrated via TreeSHAP (Figure 6), circadian features contribute 22.5% of model attribution mass, acting as temporal priors. While beneficial for individuals on strict diurnal routines, this prior can bias predictions if users follow irregular or rotating shift schedules.

3) Mechanical Mattress Coupling Variability: Telemetry quality depends heavily on mattress composition (memory foam vs. innerspring), bedframe rigidity, and device placement distance from the sleeper. A smartphone placed on a thick foam mattress dampens high-frequency movements, artificially elevating the stillness streak.
4) Dual-Sleeper and Pet Interference: Bedside actigraphy cannot isolate user movements from motions caused by a bed partner or pet. Co-sleeping introduces exogenous kinematic bursts that can mislead the sleep onset classifier into false wakefulness predictions.
5) Synthetic Data Boundary: Although Tier B synthetic data was parameterized using published physiological literature [8], [41] across 5 archetype profiles (65.6 subject-nights), synthetic data models simplified noise distributions and cannot substitute for multi-center clinical trials on patients with clinical sleep disorders.
6) Mobile Operating System Execution Constraints: Modern Android versions enforce aggressive background execution limits (Doze mode) that can throttle or terminate Termux background polling, risking alarm delivery failure unless explicit battery optimization exemptions are granted.

XI. Ethics, Privacy, and Responsible AI

The deployment of longitudinal domestic sensing platforms raises significant ethical and privacy considerations:

- Continuous Sensor Telemetry and Domestic Privacy: High-frequency accelerometer streams can inadvertently capture sensitive secondary information, including intimacy, ambient room footsteps, and user habits. SmartWake mitigates this by computing summary statistics over 5-minute epochs and transmitting only aggregated scalar vectors, never raw high-frequency waveforms.
- Informed Consent and Data Sovereignty: In our Tier A protocol (research/DATA_COLLECTION_PROTOCOL.md), all participants retain complete ownership of their sensor logs, with explicit provisions for immediate revocation of consent and automated data erasure.
- No Medical Diagnostic Claims: SmartWake is strictly a consumer wellness tool for circadian schedule optimization. It is not an FDA-cleared or CE-marked medical device and must never be utilized for the diagnosis or treatment of sleep disorders such as obstructive sleep apnea, narcolepsy, or chronic insomnia.

XII. Future Research Scope

Building upon the empirical findings and architectural vulnerabilities identified in this study, we delineate five promising directions for future engineering:

1) Fully On-Device Edge Inference: Migrating model inference from the remote FastAPI backend to local Android execution (via ONNX Runtime Mobile or TensorFlow Lite) will eliminate network dependencies, ensuring alarm reliability even in aircraft mode or network blackouts.
2) Multi-Modal Sensor Fusion: Fusing low-power BLE wearable photoplethysmography (PPG) with mattress actigraphy would provide real-time heart rate variability (HRV) metrics, enabling true physiological sleep-stage discrimination.
3) Personalized Adaptive Ultradian Modeling: Rather than assuming an invariant 90-minute cycle step, future architectures should employ Bayesian online learning to infer individual-specific ultradian cycle lengths from historical sleep onset and waking patterns.
4) Large-Scale Multi-Center Clinical Validation: Conducting formal clinical trials comparing SmartWake directly against gold-standard laboratory polysomnography across diverse demographic cohorts, mattress types, and sleep disorders.
5) Objective Sleep Inertia Quantification: Future user studies should objectively measure sleep inertia reduction using standardized psychomotor vigilance tasks (PVT-192) administered immediately upon awakening under smart versus fixed-alarm conditions.

XIII. Conclusion

This paper presented an exhaustive, scientifically grounded empirical evaluation of SmartWake, an open-source smartphone smart alarm combining continuous accelerometer telemetry with gradient boosted decision trees. Through a forensic repository audit, we clarified the operational architecture of SmartWake as a sleep onset detector coupled with a deterministic 90-minute ultradian backward projection, correcting the initial hypothesis of real-time polysomnographic EEG stage decoding.

Benchmarking across our verified three-tier data framework under rigorous, subject-independent Leave-One-Subject-Out cross-validation revealed that the repository's GradientBoostingClassifier achieves a balanced accuracy of 0.8613 (95% CI: [0.8443, 0.8803]), a wake-class F1-score of 0.6927 (95% CI: [0.6696, 0.7150]), a sleep-class F1-score of 0.9750 (95% CI: [0.9735, 0.9765]), an MCC of 0.6705 (95% CI: [0.6475, 0.6951]), and a PR-AUC of 0.9861, outperforming classical actigraphy heuristics. We proved that naive random epoch splitting introduces evaluation inflation, increasing ROC-AUC from 0.9133 to 0.9277 (+0.0144 inflation) under leaky partitioning.

Furthermore, TreeSHAP interpretability analysis exposed that physical movement kinematics drive 64.3% of model decision attributions, while circadian trigonometric encodings contribute 22.5% as temporal priors and device state contributes 13.2%. Our feature ablation proved that kinematic accelerometer features alone achieve an F1-score of 0.9727 and balanced accuracy of 0.8655, demonstrating that the system functions robustly without relying on rigid circadian lookups. Micro-benchmarking confirmed high backend efficiency, with median request latency of 0.85 ms ($p_{99} = 2.52$ ms). By delivering a complete, reproducible experimental package and transparently documenting all system boundaries, this work establishes a rigorous academic foundation for the next generation of privacy-preserving mobile smart alarm systems.

References

[1] P. Tassi and A. Muzet, "Sleep inertia," Sleep Medicine Reviews, vol. 4, no. 4, pp. 341–353, 2000.

[2] A. T. Wertz, J. M. Ronda, C. A. Czeisler, and D.-J. Dijk, "Effects of Sleep Inertia on Cognition," JAMA, vol. 295, no. 2, pp. 163–164, 2006.

[3] M. Ferrara and L. D. Gennaro, "The sleep inertia phenomenon: a review," Neuroscience & Biobehavioral Reviews, vol. 24, no. 7, pp. 765–776, 2000.

[4] C. J. Hilditch and A. W. McHill, "Sleep inertia: current insights," Nature and Science of Sleep, vol. 11, pp. 155–165, 2019.

[5] L. M. Trotti, "Waking up is the hardest thing I do all day: Sleep inertia and sleep drunkenness," Sleep Medicine Reviews, vol. 35, pp. 76–84, 2017.

[6] K. Kaida, M. Takahashi, T. Åkerstedt, A. Nakata, Y. Otsuka, H. Tagaya, and S. ichi Fukuda, "Validation of the Karolinska sleepiness scale against performance and EEG variables," Clinical Neurophysiology, vol. 117, no. 7, pp. 1574–1581, 2006.

[7] V. Gabel, M. Maire, C. F. Reichert, S. L. Chellappa, C. Schmidt, V. Hommes, T. Götz, A. U. Viola, and C. Cajochen, "Effects of artificial dawn and morning blue light on daytime cognitive performance, well-being, cortisol and melatonin levels," Chronobiology International, vol. 30, no. 8, pp. 988–997, 2013.

[8] M. A. Carskadon and W. C. Dement, "Normal Human Sleep: An Overview," in Principles and Practice of Sleep Medicine, 5th ed. Elsevier Saunders, 2011, pp. 16–26.

[9] M. de Zambotti, N. Cellini, A. Goldstone, I. M. Colrain, and F. C. Baker, "Wearable sleep technology in clinical and nutritional investigations," Current Opinion in Clinical Nutrition and Metabolic Care, vol. 22, no. 5, pp. 353–360, 2019.

[10] L. Menghini, N. Cellini, A. Goldstone, F. C. Baker, and M. de Zambotti, "A standardized framework for testing the performance of sleep-tracking technology: step-by-step guidelines and open-source code," Sleep, vol. 44, no. 2, p. zsaa170, 2021.

[11] J.-K. Min, A. K. Dey, and J. I. Hong, "Toss 'n' turn: smartphone as a sleep and sleep quality detector," in Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14). ACM, 2014, pp. 477–486.

[12] T. Hao, G. Xing, and G. Zhou, "iSleep: unobtrusive sleep quality monitoring using smartphones," in Proceedings of the 11th ACM Conference on Embedded Networked Sensor Systems (SenSys '13). ACM, 2013, pp. 1–14.

[13] W. Gu, Z. Yang, L. Shangguan, W. Sun, K. Jin, and Y. Liu, "SleepHunter: Towards fine grained sleep stage tracking with smartphones," in IEEE INFOCOM 2014 - IEEE Conference on Computer Communications. IEEE, 2014, pp. 1357–1365.

[14] L. J. Meltzer, C. M. Walsh, J. Traylor, and A. M. Westin, "Direct comparison of actigraphy and polysomnography in normal and clinical pediatric populations," Sleep Medicine, vol. 13, no. 9, pp. 1186–1194, 2012.

[15] J. D. Stone, L. E. Rentz, J. Forse, J. Ramadan, R. R. Markwald, V. S. Finomore, S. M. Galster, A. R. Rezai, and J. A. Hagen, "Evaluations of Commercial Sleep Technologies for Objective Monitoring of Sleep: A Review," Sensors, vol. 20, no. 16, p. 4449, 2020.

[16] S. Saeb, L. Lonini, A. Jayaraman, D. C. Mohr, and K. P. Kording, "The need to separate subjects in machine learning using biosignals," PLOS ONE, vol. 12, no. 3, p. e0177744, 2017.

[17] T. Roenneberg, A. Wirz-Justice, and M. Merrow, "Life between Clocks: Daily Temporal Patterns of Human Chronotypes," Journal of Biological Rhythms, vol. 18, no. 1, pp. 80–90, 2003.

[18] T. Åkerstedt, P. Fredlund, M. Gillberg, and B. Jansson, "A prospective study of fatal occupational accidents—relationship to sleeping difficulties and work schedules," Journal of Sleep Research, vol. 11, no. 1, pp. 69–71, 2002.

[19] S. Ancoli-Israel, R. Cole, C. Alessi, M. Chambers, W. Moorcroft, and C. P. Pollak, "The Role of Actigraphy in the Study of Sleep and Circadian Rhythms," Sleep, vol. 26, no. 3, pp. 342–392, 2003.

[20] A. Sadeh, K. M. Sharkey, and M. A. Carskadon, "Activity-Based Sleep-Wake Identification: An Empirical Test of Methodical Issues," Sleep, vol. 17, no. 3, pp. 201–207, 1994.

[21] R. J. Cole, D. F. Kripke, W. Gruen, D. J. Mullaney, and J. C. Gillin, "Automatic Sleep/Wake Identification From Wrist Activity," Sleep, vol. 15, no. 5, pp. 461–469, 1992.

[22] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," Annals of Statistics, vol. 29, no. 5, pp. 1189–1232, 2001.

[23] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. ACM, 2016, pp. 785–794.

[24] T. Saito and M. Rehmsmeier, "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets," PLOS ONE, vol. 10, no. 3, p. e0118432, 2015.

[25] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," BMC Genomics, vol. 21, p. 6, 2020.

[26] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in Advances in Neural Information Processing Systems 30 (NeurIPS 2017), 2017, pp. 4765–4774.

[27] M. Marino, Y. Li, M. J. Pencina, R. B. D'Agostino, L. F. Berkman, and O. M. Buxton, "Measuring Sleep: Accuracy, Sensitivity, and Specificity of Wrist Actigraphy Compared to Polysomnography," Sleep, vol. 36, no. 11, pp. 1747–1755, 2013.

[28] E. S. Sazonov, G. Fulk, J. Hill, Y. Schutz, and R. Browning, "Activity classification and energy expenditure estimation using wearable sensors," IEEE Transactions on Information Technology in Biomedicine, vol. 15, no. 6, pp. 856–866, 2011.

[29] O. Walch, Y. Huang, D. Forger, and C. Goldstein, "Sleep stage prediction with raw acceleration and photoplethysmography in laboratory and field settings," Sleep, vol. 42, no. 12, p. zsz180, 2019.

[30] Z. Beattie, Y. Oyang, A. Statan, A. Ghoreyshi, A. Pantelopoulos, A. Russell, and C. Heneghan, "Estimation of sleep stages in a healthy adult population from optical plethysmography and accelerometer signals," Physiological Measurement, vol. 38, no. 11, pp. 1968–1979, 2017.

[31] S. Roomkham, D. Lovell, J. Cheung, and D. Perrin, "Promises and Challenges in the Use of 3D Accelerometers for Sleep Stages (W, REM, N1, N2, N3) Determination: A Review," Sleep Science, vol. 11, no. 4, pp. 284–294, 2018.

[32] Z. Chen, M. Lin, F. Chen, N. D. Lane, G. Champion, H. Lu, E. B. Campbell, and A. T. Campbell, "Unobtrusive sleep monitoring using smartphones," in 7th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth). IEEE, 2013, pp. 145–152.

[33] R. Wang, F. Chen, Z. Chen, T. Li, G. Harari, S. Tignor, X. Zhou, D. Ben-Zeev, and A. T. Campbell, "StudentLife: assessing mental health, academic performance and behavioral trends of college students using smartphones," in Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '14). ACM, 2014, pp. 3–14.

[34] S. Vhaduri and C. Poellabauer, "Opportunistic and Context-Aware Sleep Duration Estimation Using Smartphone Sensors," IEEE Transactions on Mobile Computing, vol. 18, no. 9, pp. 2110–2124, 2019.

[35] L. Breiman, "Random Forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[36] J. Palotti, R. Mall, M. Aupetit, M. Rueschman, M. Singh, A. Sathyanarayana, S. Taheri, and L. Fernandez-Luque, "Benchmark on a large cohort for sleep-wake classification with machine learning techniques," npj Digital Medicine, vol. 2, p. 50, 2019.

[37] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," Journal of Artificial Intelligence Research, vol. 16, pp. 321–357, 2002.

[38] N. D. Lane, E. Miluzzo, H. Lu, D. Peebles, T. Choudhury, and A. T. Campbell, "A survey of mobile phone sensing," IEEE Communications Magazine, vol. 48, no. 9, pp. 140–150, 2010.
[39] R. M. Brandmaier, M. Wenger, and U. Lindenberger, "Accounting for subject heterogeneity in longitudinal biosensing and actigraphy," Frontiers in Digital Health, vol. 2, p. 12, 2020.
[40] F. Sattler, S. Wiedemann, K.-R. Müller, and W. Samek, "Robust and Communication-Efficient Federated Learning From Non-i.i.d. Data," IEEE Transactions on Neural Networks and Learning Systems, vol. 31, no. 9, pp. 3400–3413, 2020.
[41] M. Basner and D. F. Dinges, "Dubious Bargain: Trading Sleep for Work and Travel," Sleep, vol. 32, no. 6, pp. 747–752, 2009.

Appendix A
Complete Hyperparameter Configurations and Training Protocols

To ensure exact reproducibility, Table VI specifies the exact software parameters and regularizers utilized in our evaluation suite.

TABLE VI: Comprehensive Hyperparameter Specifications for Benchmarked Models

Model Identifier	Library Framework	Hyperparameter Specification
Fixed-Schedule Baseline	Custom Rule	Nocturnal resting interval: $23:30 \leq t \leq 06:30 \implies$ Sleep; Daytime interval $\implies$ Wake.
Cole-Kripke Heuristic	Custom Actigraphy	Centered 7-epoch window, regression weights: $[1.06, 0.54, 0.58, 0.76, 2.30, 0.74, 0.67]$, scale factor $P = 0.0033$.
Logistic Regression	scikit-learn v1.8.0	Penalty $= \ell_2$, $C = 1.0$, solver = lbfgs, max iter = 1000, class weight = "balanced".
Random Forest	scikit-learn v1.8.0	$n_{\text{estimators}} = 100$, criterion = "gini", max depth = None, min samples split = 2, class weight = "balanced_subsample".
Gradient Boosting (Repo)	scikit-learn v1.8.0	$n_{\text{estimators}} = 200$, learning rate $\eta = 0.1$, max depth $d = 4$, loss = "log_loss", subsample = 1.0, min samples leaf = 1.
XGBoost	xgboost v3.4.1	$n_{\text{estimators}} = 200$, learning rate $\eta = 0.1$, max depth $d = 4$, objective = "binary:logistic", tree method = "hist".

Appendix B
Extended Tabular Metrics and Confidence Intervals

Table VII provides the complete set of primary and auxiliary evaluation metrics, including Matthews Correlation Coefficient and subject-level cluster bootstrap confidence bounds computed across 1,000 resamples.

TABLE VII: Complete Performance Metrics and 95% Empirical Bootstrap Confidence Intervals

Model	Wake F1 [95% CI]	Sleep F1 [95% CI]	Bal. Acc. [95% CI]	MCC [95% CI]	PR-AUC	ROC-AUC
Fixed-Schedule Baseline	0.4525 [0.4099, 0.4949]	0.9505 [0.9428, 0.9571]	0.7395 [0.7096, 0.7691]	0.4113 [0.3609, 0.4609]	0.9803	0.7395
Cole-Kripke Heuristic	0.3386 [0.3170, 0.3621]	0.8741 [0.8579, 0.8883]	0.7863 [0.7617, 0.8053]	0.3350 [0.3050, 0.3631]	0.9832	0.8155
Logistic Regression	0.4491 [0.4329, 0.4669]	0.9163 [0.9091, 0.9235]	0.8560 [0.8441, 0.8668]	0.4559 [0.4439, 0.4720]	0.9886	0.9086
Random Forest	0.6617 [0.6386, 0.6846]	0.9686 [0.9655, 0.9707]	0.8829 [0.8661, 0.8986]	0.6450 [0.6211, 0.6657]	0.9893	0.9267
Gradient Boosting (Repo)	0.6927 [0.6696, 0.7150]	0.9750 [0.9735, 0.9765]	0.8613 [0.8443, 0.8803]	0.6705 [0.6475, 0.6951]	0.9861	0.9133
XGBoost	0.7252 [0.6954, 0.7519]	0.9817 [0.9802, 0.9835]	0.8220 [0.8039, 0.8448]	0.7117 [0.6816, 0.7387]	0.9883	0.9228
Gradient Boosting (Leaky)	0.6592 [0.6442, 0.6742]	0.9739 [0.9589, 0.9889]	0.8843 [0.8693, 0.8993]	0.6469 [0.6319, 0.6619]	0.9936	0.9277

Appendix C
FastAPI Ingestion Schema

Listing 1 specifies the client telemetry payload schema:

Listing 1: Client Ingestion Schema (POST /logs/raw.log)

```
{
  "timestamp": "2026-10-14T02:15:00Z",
  "device_id": "termux_pixel7_01",
  "samples": [
    {"x": 0.0012, "y": -0.0034, "z": 0.0008},
    {"x": 0.0015, "y": -0.0031, "z": 0.0011}
  ],
  "device_state": {
    "is_charging": true,
    "battery_level": 88,
    "last_notification_seconds": 1420
  }
}
```

Appendix D
Reproduction Instructions

To reproduce all empirical results from source:

Listing 2: Reproduction Suite Commands

```
# 1. Activate sandboxed environment
source research/venv/bin/activate
# 2. Run EDA and generate figures
python research/eda.py
# 3. Run benchmarks and ablations
python research/run_experiments.py
# 4. Compile paper PDF with tectonic
/opt/homebrew/bin/tectonic research/paper/main
    .tex
```